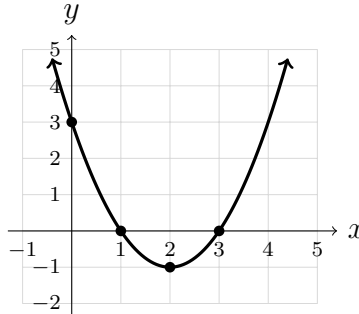


Math 133: Quiz 2 — Answer Key

Friday, September 18, 2026

1. $f(x) = (x - 2)^2 - 1 = (x - 1)(x - 3)$. Vertex: $(2, -1)$. The x -intercepts are $(1, 0)$ and $(3, 0)$; the y -intercept is $(0, 3)$. The parabola opens upward.



2. On $[0, 4]$: $f(0) = 3$, $f(2) = -1$, $f(4) = 3$. The minimum is -1 at $x = 2$; the maximum is 3 at $x = 0$ and $x = 4$.
On $[3, 4]$: $f(3) = 0$ and $f(4) = 3$. The function is increasing throughout this interval, so the minimum is 0 at $x = 3$ and the maximum is 3 at $x = 4$.
3. With capacity 4 , $q \in \{0, 1, 2, 3, 4\}$. Sell 4 baskets for a maximum profit of $P(4) = \$36$. The profit increases as q approaches 6 , so the largest allowed q gives the greatest profit.
With capacity 8 , sell 6 baskets for a maximum profit of $P(6) = \$40$. The vertex is now allowed.
4. The initial value is $\$2,000$. The factor 0.90 means the machine retains 90% of its value each year, a loss of 10% per year. $V(2) = 2000(0.90)^2 = 1620$: after two years, the predicted resale value is $\$1,620$.
5. $V(0) = 2000$, $V(1) = 1800$, and $V(2) = 1620$. The first-year loss is $2000 - 1800 = \$200$; the second-year loss is $1800 - 1620 = \$180$. The buyer is incorrect: the percentage loss is constant, but the dollar loss decreases as the machine's value decreases.